

AI-driven dermatological diagnostics: A systematic review of machine and deep learning methods for skin disease classification

Bodoor H.S. Aljohani^{*}, Wael M.S. Yafooz, Abdullah Alsaeedi

Department of Computer Science, College of Computer Science and Engineering, Taibah University, Madina, 42353, Saudi Arabia

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ABSTRACT

Skin diseases are some of the most common medical conditions worldwide, ranging from mild disorders to life-threatening cancers. Their accurate diagnosis remains challenging because of high visual variability and overlapping clinical symptoms. This systematic review, covering studies published between 2018 and 2025, evaluates computer vision, machine learning (ML), deep learning (DL), transfer learning, and hybrid approaches for dermatological disease classification. It specifically focuses on image-based (single-modal) artificial intelligence systems for dermatological diagnosis. Studies that primarily relied on textual clinical notes, electronic health records, or multimodal fusion of imaging and non-imaging data were outside the scope of this review. A total of 84 peer-reviewed studies were analyzed, with focus on datasets, model architectures, validation techniques, and reported outcomes. The review followed PRISMA guidelines and involved systematic searches across several major scientific databases. Studies were screened, assessed for quality, and systematically synthesized. The findings highlight the prevalence of convolutional neural networks (CNNs). Lightweight architectures, including MobileNet and EfficientNet variants, are frequently reported for achieving competitive accuracy with lower computational complexity, supporting their use in resource-constrained clinical settings. Transfer learning and hybrid approaches further improve adaptability and robustness across diverse datasets. Despite promising results, challenges remain in dataset diversity, fairness across skin tones, and limited clinical validation. These findings underscore the importance of responsible and deployment-aware AI design to support accurate, efficient, and equitable dermatological decision-support systems.

1. Introduction

Skin diseases are significant health problems influencing one point nine billion people worldwide and accounting for a significant number of years lived with disability (YLDs) at all ages [1]. A reliable dermatological diagnosis is still difficult owing to elevated intra-class and inter-class visual confusion, and the use of expert visual judgment creates a subjective element and shows inter-observer disputes [2]. The challenges are also being enhanced by the growing number of patients and the scarcity of trained dermatologists, especially in resource-limited environments. This has led to a lot of focus on the computer vision-based artificial intelligence systems as scalable diagnostic assistants [3]. Skin diseases are the most common disorders affecting individuals, regardless of age or race. These conditions can result from a variety of factors, including the presence of bacteria, fungi, and viruses on the skin; immune system deficiencies; allergens; irritants; genetic predisposition; and direct contact with contaminated skin. Skin diseases vary in terms of

symptoms and severity. They may take the form of acne, rashes, moles, or warts. They can be short-term or chronic and infectious, painless or painful [4]. Several are caused by environmental or temporary factors, while others are genetic. Skin diseases range from harmless issues to life-threatening diseases. While most skin problems are relatively mild, some may be a sign of a more serious health condition, and in certain cases can lead to skin cancer [5]. This means that these conditions need to be detected at the earliest to prevent their progressive existence and spread. However, diagnosing and treating skin diseases can take a lot of time, along with being both financially and physically expensive for patients [6].

A skin lesion is characteristically recognizable by its abnormally colored or textured appearance, but is difficult to diagnose with proper specificity due to overlapping clinical presentations with various diseases. For example, some benign lesions like melanocytic nevi can closely resemble malignant ones such as basal cell carcinoma (BCC) because of their similar dark colouration, making visual distinction

^{*} Corresponding author.

E-mail addresses: tu4670355@taibahu.edu.sa (B.H.S. Aljohani), wyafooz@taibahu.edu.sa (W.M.S. Yafooz), aasaedi@taibahu.edu.sa (A. Alsaeedi).

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difficult, and dermatofibroma may mimic a malignant dermatofibrosarcoma protuberans because of their comparable nodular shape and surface texture. Similarly, seborrheic keratosis and melanoma may share dark, irregular pigmentation, complicating clinical differentiation. As shown in Fig. 1, several dermatological conditions share very similar visual characteristics, making accurate differentiation challenging even for experienced clinicians [7]. This type of similarity creates unpredictability and subjectivity with clinical opinions more often, particularly for less qualified dermatologists, and hence an imperative for computerised AI-based diagnostic tools to render precise and objective opinions [8].

Based on the complexity and variability of skin diseases in terms of appearance, cause, and severity, the need for accurate and efficient diagnostic systems has become increasingly recognised [1]. Traditional diagnostic methods can be subjective, slow, and unavailable in some areas. Computing techniques, especially artificial intelligence (AI) and computer vision mechanisms, are dramatically changing the field of dermatological diagnostics. These technologies not only replicate human diagnostic capabilities but also enhance accuracy, reduce diagnostic costs, and enable real-time analysis at the point of care [6]. According to Ref. [9], computer vision focuses on creating algorithms that allow computers to understand, process, and analyze images or video to perform object recognition, image classification, segmentation, and scene reconstruction. In medical imaging, computer vision plays a crucial role in dermatological diagnostics by enabling automated image classification, segmentation, and anomaly detection [10–12].

The use of computer vision in dermatology has recently attracted significant attention from researchers due to rapid advancements in deep learning methods. Deep learning (DL) offers efficient, cost- and time-effective approaches that can be valuable in medicine [13]. Convolutional Neural Networks (CNNs) are among the DL techniques proven effective for image analysis. CNNs consist of multiple layers that apply convolutional filters to input data, allowing for automatic extraction of hierarchical features from images [14,15]. Research on CNN structures dates back to the late 1970s, when Fukushima (1980) first introduced CNNs. The application of the CNN framework to medical image analysis was initially carried out by Lo et al. (1995).

Therefore, in recent years, to create methods, models, and tools for the classification of dermatological diseases, researchers and scientists have been using ML, DL, or hybrid techniques. To improve diagnostic accuracy and effectiveness, this study aims to perform a Systematic

Review (SR) that provides a summary of the current techniques, datasets, validation strategies, and architectures suggested in academic work. Formulating research questions, establishing inclusion and exclusion criteria, locating electronic databases, selecting studies, extracting data, and synthesising findings are the primary steps in conducting this systematic review.

This study answers four Research Questions (RQs).

- **RQ1:** Which methodological trends (machine learning, deep learning, transfer learning, and hybrid models) are typical of dermatological skin disease classification research in the articles published between 2018 and 2025?
- **RQ2:** What datasets have been used in the dermatological disease classification research, and what are their primary features in terms of size, distribution and usability?
- **RQ3:** Which validation methods and measures tend to be applied to evaluate the performance of AI-based dermatological disease classification models?
- **RQ4:** What is the performance of lightweight architectures, especially MobileNet and its variants, compared to other deep learning architectures in the cases of diagnostic accuracy and computation efficiency?

The structure of this paper is organised as follows: Section 2 explains the main terminologies and definitions used in this SR. Section 3 describes the materials and methods, including the search strategy, study selection, data extraction and synthesis, and quality assessment. Section 4 presents the results of the review, covering study characteristics, datasets, models, and techniques used. Section 5 discusses the findings, comparative analysis, and implications. Section 6 concludes the paper with key remarks and recommendations for future research.

2. Terminologies and definitions

This section explains the main terminologies and definitions used in this SR that are relevant to the research area. Further, it illustrates the impact of dermatological disease classification using artificial intelligence. These are some of the terminologies used in this SR. These terms are defined below.



Fig. 1. Clinical visual similarity among common dermatological conditions, illustrating diagnostic challenges.

2.1 Dermatological Disease: Dermatological diseases are conditions that affect the skin, ranging from mild disorders such as acne and rashes to severe and life-threatening conditions like melanoma. They may be caused by genetic, environmental, or infectious factors, and represent a significant global health burden [3,4].

2.2 Machine Learning (ML) is a subset of AI that enables systems to learn patterns from data and improve their performance over time without explicit programming. Traditional ML techniques such as Support Vector Machine (SVM) and Random Forest (RF) were initially used for skin disease classification, relying heavily on handcrafted features [16]. The general workflow of a ML classification system is explained in Fig. 2, highlighting the stages of feature extraction, feature selection, and classification [1].

2.3 Deep Learning (DL) is a specialised branch of ML that uses multi-layered neural networks to automatically extract features from input data. It has proven highly effective in medical image analysis, including skin disease classification [17]. As shown in Fig. 3, CNNs automatically learn hierarchical features from input dermoscopic images during convolution, pooling, and fully connected layers [1].

2.4 Hybrid Models: Hybrid models combine two or more techniques—such as ML and DL methods—to improve accuracy and generalisation. For example, CNNs can be used for feature extraction, while ML classifiers like SVMs can perform the final classification [8].

2.5 Convolutional Neural Network: CNNs are a specialised architecture for DL designed for photographs. CNNs utilise convolutional filters to help CNNs learn hierarchical image features. Proof of this is their most extensive use in dermatological image classification [17].

2.6 MobileNet: MobileNet is a lightweight CNN architecture designed for mobile and lower resource situations. It achieves good diagnostic accuracy with lower computational requirements, making it effective for real-time classification of skin diseases [13].

2.7 Transfer Learning: This is used when adapting models already trained on data (like ImageNet) to new domains. For example, adapting a model to classify dermatological diseases. This is a cost-effective method with better results, especially when the annotated medical datasets are small [10,18].

2.8 Lightweight Architectures: Lightweight architectures are DL models designed to achieve computational complexity with an accuracy pedestal. These include models such as MobileNet, EfficientNet, and SqueezeNet, which are used to deploy AI-based diagnostic tools in clinical and mobile applications.

3. Methods

This section presents the research methodology used to conduct a systematic review of methods and approaches applied in dermatological disease classification using DL. It contains four main phases. They are the strategy of search, study selection, data extraction, and analysis of results, as shown in Fig. 4.

3.1. Search strategy

This is a systematic review that was carried out as per the PRISMA 2020 guidelines. A detailed literature search was conducted in seven electronic databases: IEEE Xplore, PubMed, MDPI, ScienceDirect,

SpringerLink, ACM Digital Library, and Google Scholar. The search was conducted between 14 April 2025 and 25 October 2025, covering studies published between January 2018 and October 2025. It utilised a concept-based Boolean search strategy that involves three essential elements: (i) dermatological conditions, (ii) artificial intelligence methods and (iii) image-based analysis. Maximising coverage and relevance was achieved using Boolean operators (AND/OR). Peer-reviewed publications in the English language were further limited by database-specific filters.

The search process began with exploratory keyword-based queries (e.g., “skin disease classification,” “dermatology classification,” “deep learning in skin disease detection,” and “hybrid model for skin disease detection”) to identify commonly used terminology in the field. As the review progressed, structured Boolean combinations using AND/OR operators were applied to refine and narrow the results. Search terms combined concepts related to dermatological conditions (e.g., “skin disease,” “skin lesion”), artificial intelligence techniques (e.g., “deep learning,” “machine learning,” “convolutional neural network,” “transfer learning,” “hybrid model”), and imaging context (e.g., “dermoscopy,” “medical imaging”). Filters were applied to limit results to English-language peer-reviewed publications published between January 2018 and October 2025. The complete verbatim search strings for all databases, including database-specific syntax and field tags, are provided in Appendix A. All retrieved records were exported into a reference management tool, where duplicate entries were automatically removed and manually verified.

3.2. Study selection

The selection of studies was done in a two-stage screening. The process of screening titles and abstracts in the first stage helped to filter out studies which were obviously not relevant to the classification of dermatological diseases or that used methods other than image-based artificial intelligence. The second stage included predefined inclusion and exclusion criteria that were used to screen the full-text articles on eligibility. Inconsistencies in the screening phase were addressed by discussing and reducing selection bias. Any disagreements were resolved through discussion until consensus was achieved. All predefined inclusion and exclusion criteria were systematically applied during the screening process. The complete study selection workflow, including identification, duplicate removal, screening, eligibility, and final inclusion, is illustrated in the PRISMA flow diagram (Fig. 4). A total of 287 records were identified through database searching. After removing duplicates ($n = 17$), 270 records remained for screening. Of these, 270 records were screened based on titles and abstracts, and 161 full-text articles were assessed for eligibility. The reasons for full-text exclusion along with the corresponding numbers of articles excluded on each criterion were as follows: not human dermatological disease classification ($n = 17$), review or survey articles ($n = 14$), lack of quantitative performance metrics ($n = 15$), non-image-based approaches ($n = 11$), and studies outside the defined methodological scope ($n = 20$). Finally, 84 studies were included in the systematic review. Table 1 summarizes the number of studies identified and those selected from each database.

As shown in Table 1, Google Scholar and IEEE Xplore contributed the largest number of relevant studies, while other databases provided smaller but complementary sets of articles. Detailed counts of duplicate

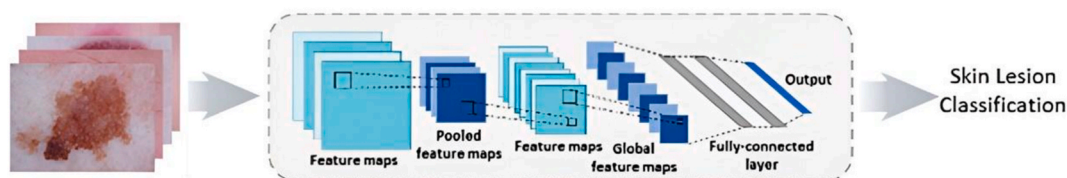


Fig. 2. General workflow of a machine learning-based skin disease classification system.

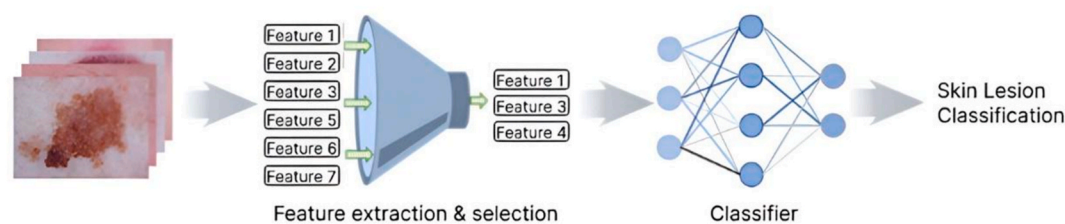


Fig. 3. Deep learning workflow for skin disease classification using convolutional neural networks.

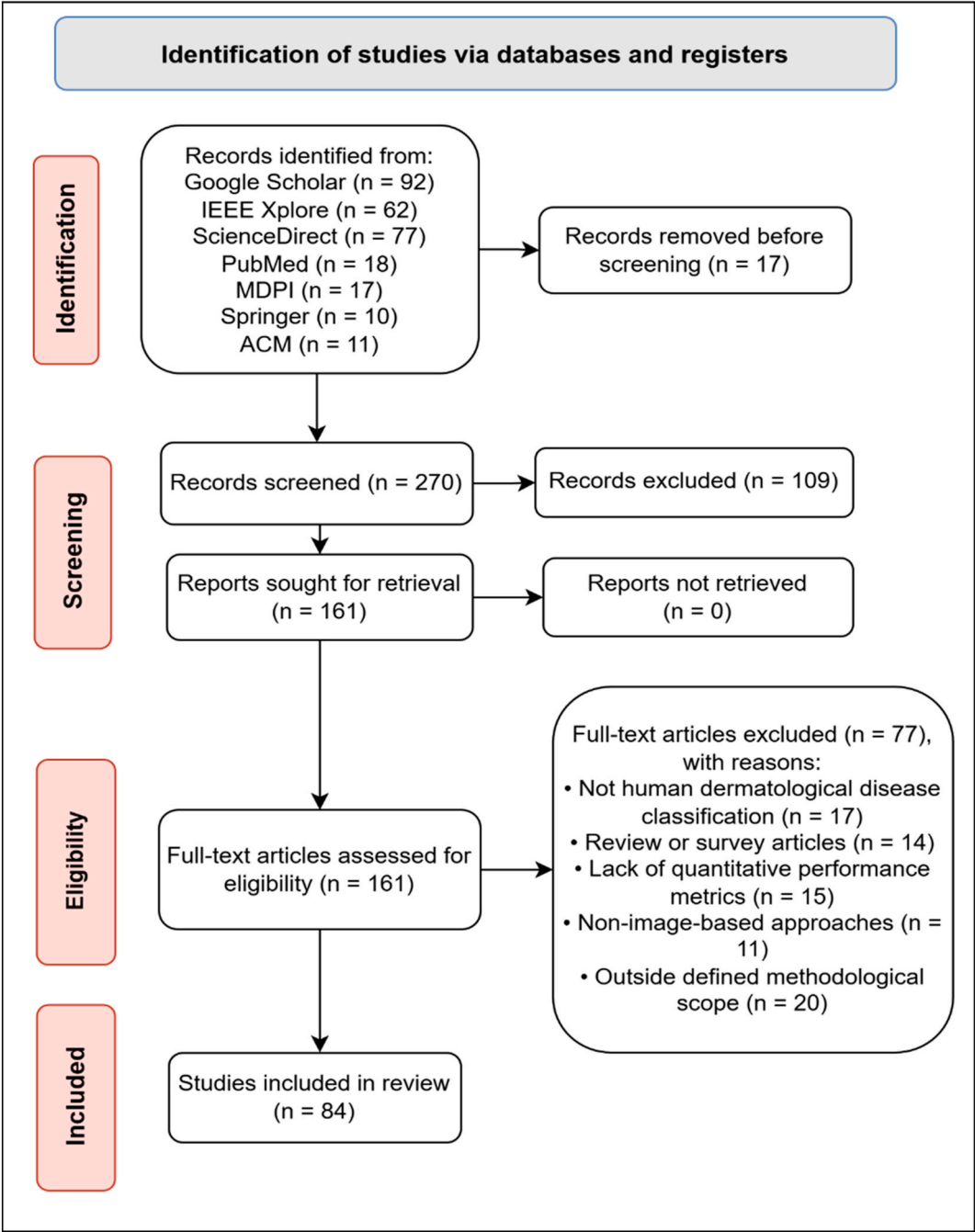


Fig. 4. PRISMA 2020 flow diagram illustrating the study selection process.

removal, screening exclusions, and full-text eligibility decisions are provided in the PRISMA flow diagram (Fig. 4) and summarised in the text above.

Table 1
Search results across databases.

Database	Found	Selected
Google Scholar	92	19
IEEE Xplore	62	46
ScienceDirect	77	5
PubMed	18	5
MDPI	17	2
Springer	10	5
ACM	11	2

3.3. Inclusion and exclusion criteria

- 3.3.1 Inclusion Criteria:
- 3.3.1.1 Peer-reviewed journal articles and conference proceedings.
 - 3.3.1.2 English-language publications.
 - 3.3.1.3 Studies published between January 2018 and October 2025.
 - 3.3.1.4 Studies employing artificial intelligence methods (ML, DL, transfer learning, or hybrid models) for dermatological disease classification.
 - 3.3.1.5 Studies reporting quantitative evaluation metrics (e.g., Accuracy, AUC, F1-score).
 - 3.3.1.6 Studies employing image-only dermatological datasets (clinical or dermoscopic images) as the primary input modality.
- 3.3.2 Exclusion Criteria
- 3.3.2.1 Non-English publications.
 - 3.3.2.2 Articles published before 2018.
 - 3.3.2.3 Studies unrelated to dermatological disease classification.
 - 3.3.2.4 Books, theses, dissertations, and non-peer-reviewed works.
 - 3.3.2.5 Studies lacking quantitative evaluation.
 - 3.3.2.6 Studies primarily based on multimodal data fusion approaches (image combined with clinical text, electronic health records, or laboratory findings).

The inclusion and exclusion criteria applied during the study selection process are summarised in Table 2 to ensure transparency and reproducibility (see Table 3).

3.4. Data extraction and synthesis

The data extraction process involved a systematic review of selected articles to capture key information relevant to this study. Extracted data included: (1) methodological approach (e.g. ML, DL, transfer learning, hybrid model); (2) author(s) and year of publication; (3) dataset used including dataset name, size, and source; (4) number of disease classes; (5) model architectures or classifiers applied (e.g., MobileNet, EfficientNet, ResNet, (SVM)); and (6) the reported performance metrics (accuracy, F1 score, (AUC)), as well as any stated limitations. This structured synthesis ensured consistency between studies and facilitated

Table 2
Study selection criteria.

Aspect	Inclusion Criteria	Exclusion Criteria
Publication type	Peer-reviewed journal articles and conference papers	Reviews, theses, editorials, books, non-peer-reviewed works
Language	English	Non-English
Publication period	2018–2025	Published before 2018
Study focus	Dermatological disease classification	Unrelated to dermatology or classification
Methodology	AI-based methods (ML, DL, TL, Hybrid)	Non-AI approaches
Input modality	Image-only (single-modal) data	Multimodal data (image + text/tabular)
Evaluation	Quantitative metrics (Accuracy, F1, AUC)	No quantitative evaluation

Table 3
Characteristics of included studies.

Feature	Distribution
Publication Years	2018–2025 (majority after 2020)
Publication Sources	IEEE Xplore, PubMed, Web of Science, Springer, ACM, Google Scholar
Geographic Regions	India, China, USA (majority); Europe, Middle East (smaller contributions)
Approaches Used	ML, DL, Hybrid Approaches
Classification Scope	Binary (benign vs malignant) and multi-class (7–500+ classes)
Reported Metrics	Accuracy, F1 score, Sensitivity, Specificity, (AUC)

the identification of trends, strengths, and research gaps in dermatological disease classification using AI. In addition to the extracted variables, two methodological indicators were included in the performance tables: (1) external validation, which was reported as Yes/No, and (2) data leakage safeguards, which were reported as Yes/No/Not reported. External validation was defined as the evaluation of the models on a separate dataset not used during the training process. Data leakage safeguards referred to the reported use of specific strategies to avoid data overlap between the training set and the test set, such as the use of patient-level separation or duplication removal.

3.5. Quality assessment

To ensure the reliability and validity of this systematic review, the methodology for evaluating the quality of the included studies was carefully assessed. Each selected article was evaluated according to the eligibility criteria described in the 3.1 Search Strategy section. The main aspects considered were.

- 3.5.1 Peer-reviewed publication: Only studies published in indexed journals or conference proceedings were included, while preprints and non-peer-reviewed works were excluded to maintain scientific rigour.
- 3.5.2 Clarity of research design: Studies were assessed based on the transparency of their methodology, including the description of datasets, preprocessing techniques, model architectures, and validation strategies.
- 3.5.3 Reporting of performance metrics: A key requirement was the presentation of quantitative evaluation measures such as accuracy, F1 score, sensitivity, specificity, or AUC. Articles lacking clear metrics or providing only qualitative descriptions were excluded.
- 3.5.4 Reproducibility: Priority was given to studies that employed publicly available datasets or clearly defined data collection procedures, as well as those that specified hyperparameters and training strategies.
- 3.5.5 Relevance to scope: Only articles directly addressing dermatological disease classification using AI techniques, particularly ML and DL, were included. General AI papers without medical application, or those outside dermatology, were excluded.

This multi-criteria evaluation ensured that the final pool of studies not only reflected the breadth of AI applications in dermatology but also maintained a consistent level of methodological rigour. By filtering out studies with insufficient detail or unclear results, the review focused on research that offers reliable evidence and meaningful insights for clinical and technological advancement.

3.6. Methodological quality appraisal of included studies

Because the included studies varied substantially in design, datasets, and evaluation settings, a single formal risk-of-bias tool was not applied uniformly. Instead, a structured methodological quality appraisal was conducted to assess the reliability and interpretability of the included

image-based AI studies. The appraisal considered dataset representativeness and diversity, annotation and reference-standard clarity, transparency of preprocessing and model development, validation rigour, and completeness of performance reporting. The heatmap in Fig. 5 illustrates the performance of each of the evaluated studies across five key methodological areas. Studies with large, diverse datasets and well-defined annotation or reference standards are represented by lighter cells in the first two columns. Conversely, studies with smaller,

unbalanced, or poorly documented datasets show darker cells. Furthermore, studies that clearly explain their preprocessing steps, model development, effective validation strategies (including external validation), and detailed reporting of performance metrics tend to have more light-colored cells in the remaining columns. The overall pattern of darker cells across most studies highlights recurring issues identified in the review, such as dataset imbalance, limited diversity in skin tones, and absence of external validation. Addressing these concerns should be

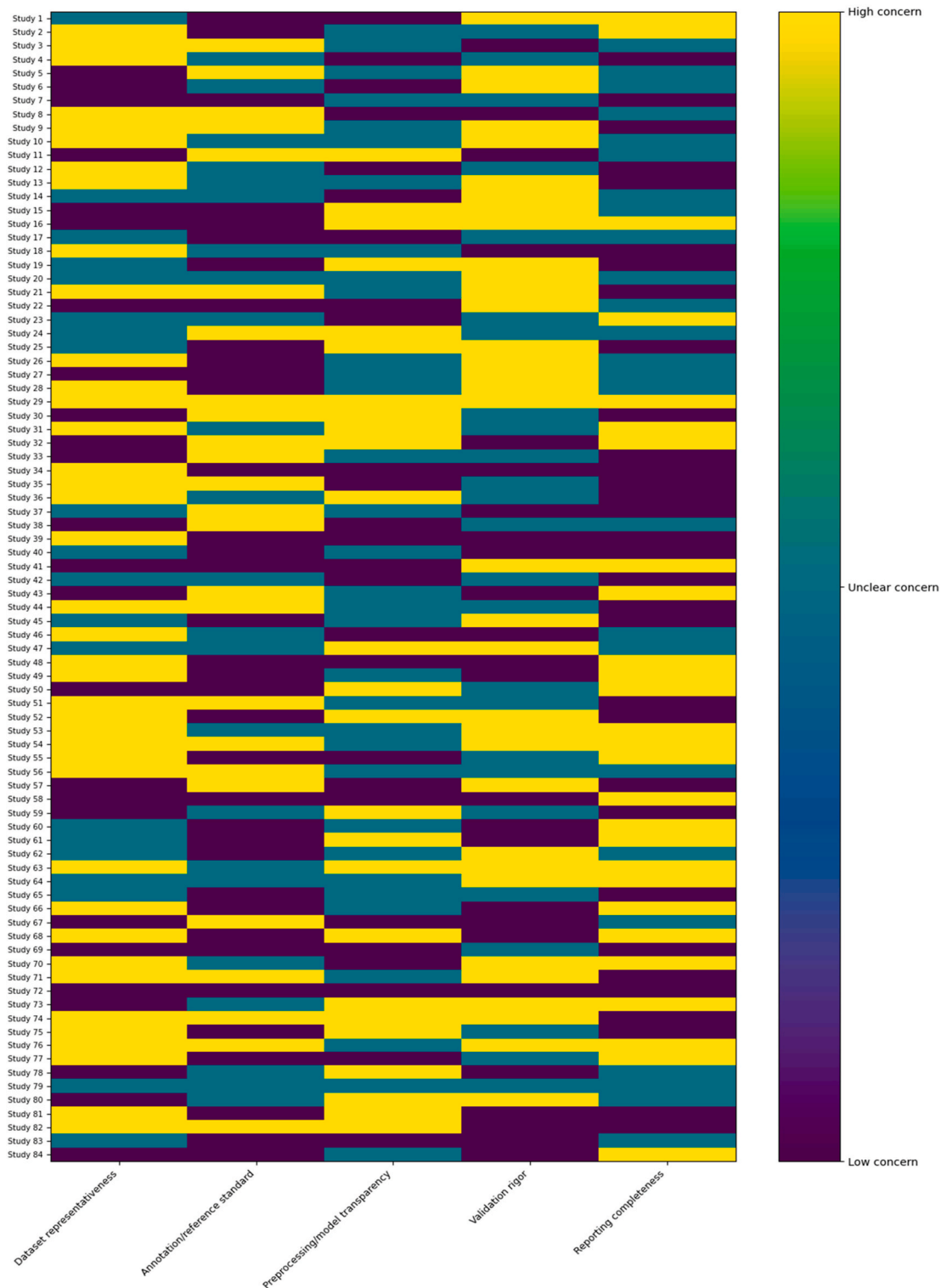


Fig. 5. Per-study methodological quality/credibility heatmap across included studies using adapted QUADAS-2/AI-informed domains.

a priority for future research to strengthen the evidence base.

As summarised in Fig. 6, common methodological concerns included dataset imbalance, limited skin-tone diversity, lack of external validation, and inconsistencies in evaluation protocols.

Studies with clearer dataset documentation, stronger validation procedures, and more transparent reporting were given greater interpretive weight in the narrative synthesis. Although many studies reported promising performance, the evidence base remains limited by recurring concerns related to diversity, fairness, reproducibility, and clinical validation.

4. Results

In the final stage of the review process, there were 84 relevant research articles in all. These articles were divided into four primary categories: ML, DL, Transfer Learning, and hybrid approaches. Within the DL group, further categorisation was made to distinguish lightweight models such as MobileNet from other CNN-based architectures.

The research articles were then summarised based on several criteria. First, the study's main objective. Second, the dataset employed, including its size and number of disease classes. Third, the classifiers or DL models that were applied (e.g., MobileNet, EfficientNet, ResNet, SVM). Fourth, the language or context of the dataset. Fifth, the reported performance metrics, such as accuracy, F1 score, and AUC. At the end of each group, a critical analysis was highlighted, and future perspectives for research were outlined.

4.1. Study characteristics

The reviewed studies are explicitly categorised into three task types: classification, segmentation, and detection. Performance results are discussed strictly within each task category using task-appropriate evaluation metrics. The systematic review included 84 research articles published between 2018 and 2025, covering a wide range of approaches in dermatological disease classification. The majority of studies were published in the last five years, reflecting the growing interest in applying DL and lightweight architectures such as MobileNet for medical imaging tasks. Most of the included studies were conference papers and journal articles indexed in databases such as IEEE Xplore, PubMed, Springer, and Web of Science.

In terms of geographical distribution, the studies originated from multiple regions, with a noticeable concentration in India, China, and the United States, where large dermatological datasets and strong AI research communities are available. A smaller but meaningful number of studies came from Europe and the Middle East, often relying on publicly available datasets rather than locally collected images. As illustrated in Fig. 7, the studies are distributed across multiple countries, with India leading, followed by China and Bangladesh, and additional contributions from the United States, Europe, and the Middle East see Fig. 7.

The model categories are identified in this review into four primary

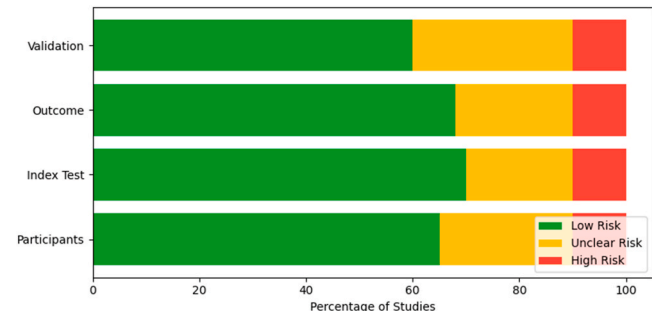


Fig. 6. Methodological quality summary across included studies.

categories: ML, DL, Transfer Learning, and Hybrid Approaches. DL-based studies accounted for the largest share, reflecting the dominance of CNN-based architectures in medical image classification tasks. Hybrid methods, although fewer in number, employed a combination of ML and DL techniques to enhance classification accuracy and generalization.

In the literature, several studies focus on binary classification (e.g., benign vs malignant), while others address multi-class classification involving 7 classes or more than 500 classes of dermatological conditions. These studies reported the performance of their approaches based on accuracy, F1 score, sensitivity, specificity, and AUC; however, reporting was not consistent across all studies. Table 3 summarizes the key characteristics of the included studies.

4.2. Dataset characteristics

In dermatological research studies, different datasets have been employed to train and test ML and DL models. While several datasets are publicly obtainable and extensively used, other datasets are institution-specific or private and available only to authorised researchers. Table 4 provide a major of the datasets, including their size, number of disease classes, image size, and obtainability.

The reviewed studies relied heavily on publicly available dermatology datasets, with ISIC and HAM10000 being the most frequently used due to their accessibility and standardized dermoscopic images. Furthermore, large-scale datasets such as Xiangya Derm (over 100,000 images) provide extensive coverage of diverse diseases, but access restrictions limit their use. In contrast, smaller datasets such as DermoFit and DDI highlight the persistent challenge of dataset imbalance and limited disease diversity, which can negatively impact model generalization. The Fitzpatrick17k dataset is particularly valuable because it includes annotations across different skin tones, addressing concerns about fairness and bias in dermatology AI research. Similarly, the DERM12345 dataset is one of the newest collections captured using high-resolution digital cameras and dermatoscopes under clinical supervision, providing diverse and clinically validated images that reflect real-world dermatological conditions [7].

4.3. Models and techniques used

Classification is the primary task in ML and AI, where the objective is to classify input data, such as images, text, and signals, into predefined categories/classes. In the domain of skin disease diagnosis, classification algorithms can automatically categorize medical images based on their dermatological classification into categories. Effective classification support results in correct early detection, accurate diagnosis, and optimized treatment. Thus helping patients and reducing pressure on health professionals [16].

Classification algorithms are commonly categorised into traditional ML classifiers, which depend on hand-crafted properties and traditional statistical techniques, and DL classifiers, which automatically learn hierarchical representations from raw data, especially successful for image-based tasks. In recent studies, these categories have further expanded to include hybrid approaches that combine ML and DL, and transfer learning approaches that leverage pre-trained CNNs such as VGG, Inception, and MobileNet. An overview of these four approaches is presented in Fig. 8.

4.3.1. Machine learning classifiers (ML)

The initial research on the classification of skin diseases primarily utilised conventional ML classifiers, including SVM, k-NN, Random Forest, Naive Bayes, and ANN in conjunction with hand-engineered features (e.g., color, texture, and shape). For example [27], extracted RGB and GLCM features and four different classifiers, LDA produced 82.58% accuracy using the color feature, while SVM provided the best performance using the texture feature set. KNN and Random Forest were

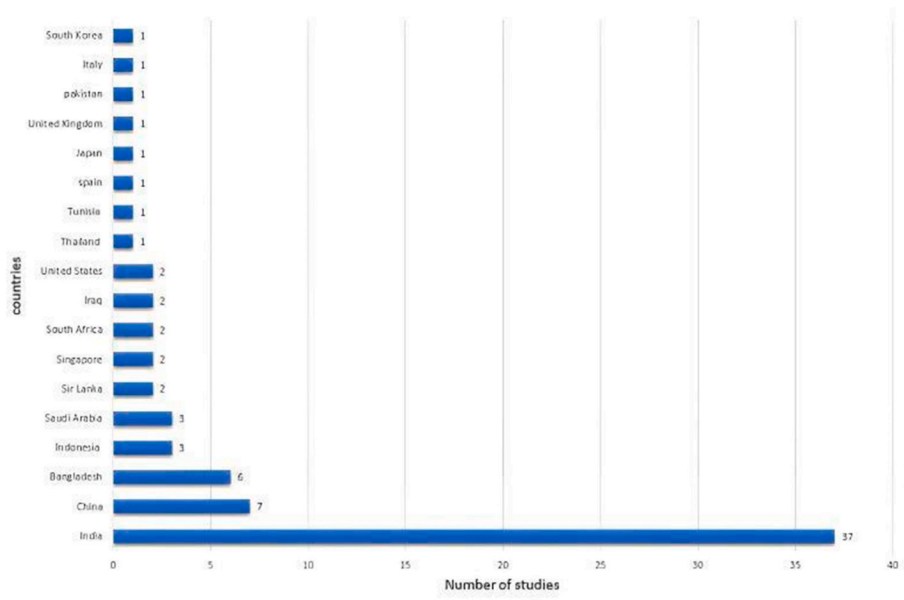


Fig. 7. Geographic distribution of the studies included in this systematic review (2018–2025).

Table 4
Summary of publicly available dermatology datasets.

Dataset	Year	Number of Images	Classes	Modality	Availability/ License	Skin-tone Labels	Image Dimensions	Ref
DermNet	1998	23,000	23 diseases	Clinical	Restricted	No	540 × 722 to 4499 × 6748	[19]
DermoFit	2013	1300	10 diseases	Clinical	Restricted	N/A	380 × 380	[20]
PH2	2013	200	3 classes	Dermoscopic	Public	Limited (Fitzpatrick II–III only)	766*576	[21]
ISIC Archives	2016–2020	2094–33,126	9 diseases	Dermoscopic	Public	Not Provided	Variable	[22]
HAM10000	2018	10,015	7 diseases	Dermoscopic	Public	Not Provided	400 × 600	[23]
Xiangya Derm	2019	107,565	541 diseases	Clinical	Public	Asian cohort; no skin-tone labels	Variable	[24]
DDI	2022	656	Benign/ Malignant	Clinical	Public	Yes (Fitzpatrick skin types I–VI)	224 × 224	[25]
Fitzpatrick17k	2022	16,577	114 subclasses	Clinical	Public	Yes (Fitzpatrick I–VI)	Variable	[26]
DERM12345	2024	12,345	40 subclasses	Dermoscopic	Public	Not Provided	3840 × 2160	[7]

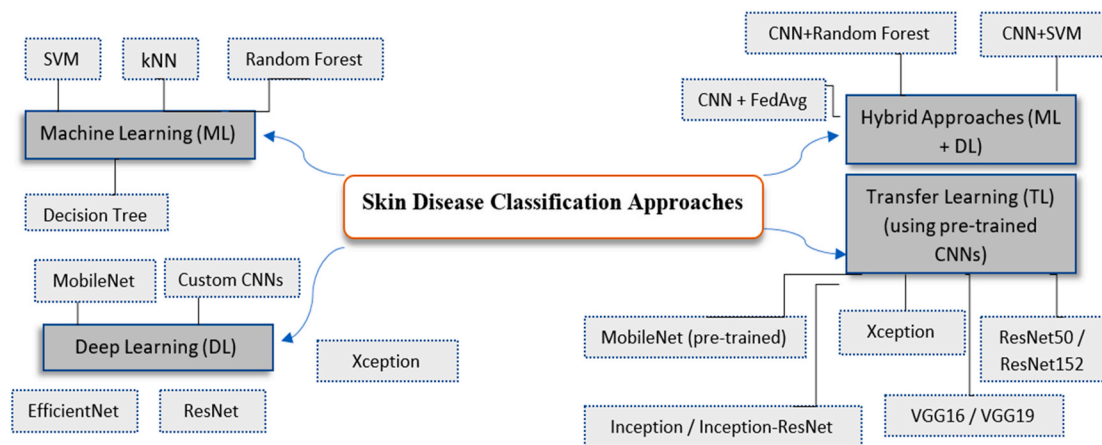


Fig. 8. Overview of methods and techniques for dermatological disease classification.

tested using a dataset consisting of 21,075 images distributed among ten classes, and gave high performance (KNN: 95.23%, RF: 94.22%) [28]. In several research that integrating ML algorithms as part of an ensemble model yielded better performance and achieved an accuracy of 97.33% using the dataset provided by the authors and consisting of five classes

for the classification of skin diseases [29]. In study [30], the authors used the DermNet dataset containing 733 images classified into four classes (Acne, Cherry angioma, Melanoma, and Psoriasis), using SVM, RF, and k-NN classifiers, where the SVM achieved the highest accuracy of 90.7%. In study [31], the ISIC 2017 dataset with 1646 images across

three classes (Benign, Melanoma, and Keratosis) was used, and a Dynamic Graph Cut combined with a Naïve Bayes classifier achieved an accuracy between 91% and 94.3%. Similarly, in study [32], authors used the ISIC dataset (1000 images, two classes: Benign and Melanoma) using SVM and k-NN, which resulted in an accuracy of 89.43%. These studies collectively demonstrate the capability of traditional ML models for skin disease classification. Although such findings are promising, ML classifiers remain feature-engineering-intensive and often exhibit limited generalizability compared with DL models.

Table 5 summarizes selected studies that applied ML classifiers for skin disease classification. It highlights the dataset sources, number of images, disease classes, applied algorithms, and the reported performance results.

The results from studies employing traditional ML classifiers indicate that these methods can achieve relatively high accuracy when applied to dermatological image datasets. For example, ensemble models combining SVM, KNN, RF, and NB, the accuracies become above 97% [29], while simpler models such as KNN and RF also produced high results exceeding 95% on larger benchmark datasets [28]. However, the performance of some classifiers, such as LDA and Naive Bayes, was more modest, with accuracies around 82% [27,32], reflecting their limited ability to generalize across diverse skin conditions. Overall, ML approaches demonstrate potential for skin disease classification, but their reliance on handcrafted features and limited generalization compared with DL models reduces their applicability for complex, large-scale medical imaging tasks.

4.3.2. Deep learning classifiers (DL)

DL is a branch of AI that has rapidly become transformative for healthcare, with the ability to analyze data at speed. DL is a type of ML that uses a multi-layer neural network to analyze data [36]. One type of DL, known as CNN, has emerged as a highly effective approach for the classification and diagnosis of skin diseases, showing significant potential to enhance speed and accuracy in medical technology [36–43]. CNNs are adept at automatically extracting hierarchical features from complex images, which is crucial for distinguishing between visually similar skin conditions [43,44]. Several well-known CNN architectures and customized models have been employed and enhanced for this purpose.

Common CNN architectures. Recent studies have explored variants of CNN architectures for dermatological image classification. ResNet and VGG16 have been widely adopted in skin disease classification. Modified ResNet50 architectures, which combine additional fully connected layers and dropout regularization, have achieved high accuracy, such as 98.93% for classifying eight types of skin diseases [37]. Similarly, VGG16 has demonstrated an accuracy of 92.8% in multi-class cancer

detection [45,46].

On the other hand, MobileNet is a lightweight CNN architecture suitable for mobile applications, achieving competitive accuracy (up to 88% with slight modifications) [5,40]. Enhanced variants such as MobileNetV2 with LSTM and MobileNet-RseSK, which improved performance on dermatological datasets like HAM10000 [36,39].

SqueezeNet is a lightweight model that prioritizes accuracy with a significantly reduced number of parameters. A modified version of SqueezeNet achieved 97.04% accuracy on the HAM10000 dataset [44].

Table 6 summarizes selected studies that applied DL classifiers trained from scratch (non-transfer learning) for skin disease classification. It highlights the dataset sources, number of images, disease classes, applied algorithms, and the reported performance results.

DL classifiers, especially CNN-based architectures, demonstrate strong performance across a variety of dermatological datasets. Moreover, classic models like ResNet50 and VGG16 achieved accuracies above 90% in several multi-class classification tasks [49]. Lightweight CNNs, such as MobileNet and its modified versions like MobileNet-RseSK, generally produced competitive results, ranging between 85% and 95% while maintaining lower computational requirements [36,39]. Advanced architectures such as EfficientNet and modified ResNet50 reached peak accuracies close to 99% in specific datasets [37,43], showing the impact of architectural optimization and fine-tuning. Nevertheless, performance varied significantly depending on dataset size and diversity: small or institution-specific datasets led to lower generalisation, e.g., 80% on a small custom dataset [47], while large-scale datasets like HAM10000 and ISIC enabled models to reach state-of-the-art performance. In a study by Sakli et al., a lightweight CNN (LWCNN) model trained on the HAM10000 dataset achieved an accuracy of 99.34% [42]. These results demonstrate the potential of optimized CNN-based architectures to reach near-perfect classification performance when trained on well-curated dermatological datasets. Overall, DL classifiers clearly outperform traditional ML methods in terms of scalability and feature extraction, but their dependence on large, balanced datasets remains a major limitation. Table 7 reviews current deep learning research that centres on skin disease detection and segmentation, with models being assessed by task-relevant measures like the localization accuracy, mAP, Dice coefficient, and the IoU. These studies are a complement to classification-based methods and allow independent lesion localization and boundary delineation, which are essential in clinical interpretation and automated diagnosis.

4.3.3. Transfer learning classifiers

The Transfer Learning relies on a pre-training procedure on a large dataset, on a large dataset such as ImageNet, and is then fine-tuned to suit the requirements of a specific new task. The top layer of the pre-trained model is replaced with a new layer, and the training process is

Table 5
Summary of selected studies using ML classifiers for skin disease classification.

Ref	Year	Dataset	Dataset Size	Classes	Classifiers	Result	External validation	Leakage safeguards
[29]	2023	Kaggle	750 images	5 classes	(SVM), (KNN), RF, Naïve Bayes (NB)	97.33%	N	Not Reported
[27]	2018	Private (Yenepoya Medical College, India)	157 images	3 classes	LDA, Naive Bayes, (ANN), (SVM)	LDA (Color): 82.58%	N	Not Reported
[6]	2024	PH ²	200 images	2 classes	ANN, SVM	94%	N	Not Reported
[33]	2019	N/A	45 images	4 classes	SVM	96–98%	N	Not Reported
[28]	2021	Open source benchmark dataset	21,075 images	10 classes	(KNN), (RF)	(KNN): 95.23%	N	Not Reported
[30]	2024	Kaggle	300 images	3 classes	Decision Tree (SVM)	(SVM): 96.1%	N	Not Reported
[34]	2022	DermNet	733	4 classes	SVM, RF, K-NN	SVM: 90.7%	N	Not Reported
[31]	2020	ISIC 2017	1646	3 classes	Dynamic Graph Cut + Naïve Bayes	91–94.3%	N	Not Reported
[32]	2019	ISIC	1000	2 classes	SVM, k-NN	89.43%	N	Not Reported
[35]	2025	Histopathological biopsy	N/A	Multi-class	RF, LR, SGD, SVM, AdaBoost, NB	SGD = 99.09%	N	Not Reported

Table 6

Summary of selected deep learning studies for skin disease classification.

Ref	Year	Dataset	Dataset Size	Classes	Classifier(s)	Model Efficiency	Result	External validation	Leakage safeguards
[47]	2022	Custom dataset	30 images	malignant and benign	CNN	N/A	80%	N	Not Reported
[5]	2024	Kaggle	900 images	9 classes	Custom CNN, GoogLeNet, MobileNet	N/A	MobileNet: 85%	N	Not Reported
[48]	2023	Published datasets	1300 images	2 (Light, Dark skin)	Modified MobileNet	≈4.2M	97%	N	Not Reported
[49]	2019	Xiangya Derm	4394 images	6 classes	ResNet50, InceptionV3, DenseNet121, Xception, Inception-ResNet-v2	N/A	92.9%	N	Not Reported
[38]	2023	Kaggle	6286 images	2 (Eczema, Psoriasis)	Custom CNN (15 layers)	2 M	96.2%	N	Not Reported
[36]	2020	HAM10000	10,015 images	7 classes	MobileNet	N/A	Val: 72%, Melanoma: 95%	N	Not Reported
[37]	2024	Online sources	700 images	8 classes	Modified ResNet50	N/A	98.93%	N	Not Reported
[44]	2023	HAM10000	10,015 images	7 classes	Modified SqueezeNet	0.13M	97.04%	N	Not Reported
[50]	2025	ISIC2019 + HAM10000	35,346 images	15 classes	Custom CNN	N/A	88.2%	N	Not Reported
[51]	2024	ISIC	2800 images	8 classes	Customized CNN	0.218M	95%	N	Not Reported
[40]	2022	Kaggle	21,485 images	10 classes	Custom-built CNN	N/A	91.07%	N	Not Reported
[52]	2024	DermNet + HAM10000	7339 images	5 classes	AlexNet, VGG19	N/A	96.83%	N	Not Reported
[39]	2024	HAM10000	10,015 images	7 classes	MobileNet-RseSK	N/A	86.5%	N	Not Reported
[42]	2023	HAM10000	10,015 images	8 classes	LWCNN	1.612 M	99.34%	N	Not Reported
[53]	2019	ISIC 2017	2750 images	3 classes	Hierarchical CNN (DenseNet-161)	N/A	BACC: 70%	N	Not Reported
[54]	2025	Custom dataset	N/A	4 classes	EfficientNet	N/A	96.7%	N	Not Reported
[55]	2024	Monkeypox + DermNet	N/A	N/A	MobileNetV2	3.7 M	92.28%	N	Not Reported

Table 7

Summary of deep learning studies for skin disease detection and segmentation.

Ref	Year	Dataset	Dataset Size	Task Type	Model	Evaluation Note
[56]	2023	Kaggle	400 images	Detection + Classification	CNN + YOLOv8	Detection performance reported using localization metrics; classification accuracy 85%
[57]	2024	ISIC + Kaggle + Private	>1000 images	Detection	YOLOv8	Detection performance reported (mAP/localization)
[46]	2024	Online sources	6000 images	Detection + Classification	YOLOv3, VGG19, MobileNet	VGG19 classification: 92.8%; detection evaluated separately
[58]	2024	DermNet + HAM10000	1012 images	Segmentation + Classification	U-Net CNN, VGG19, MobileNetV2	Segmentation evaluated using pixel-level metrics; classification: 93%
[59]	2020	N/A	600 images	Detection + Classification	Faster R-CNN, MobileNetV2	Detection performance: 87.2%; classification evaluated separately

repeated on the new dataset for the required task. This allows a model originally trained for one problem to work for another model addressing a different problem. This helps the model to learn faster as well as reduce the number of computations [60]. Transfer learning classifiers are applied to dermatological image classification through pre-trained CNN architectures such as VGG, ResNet, and MobileNet families.

VGG architecture (VGG16, VGG19) is widely used and was able to classify 1000 image classes with up to 92.7% accuracy when trained on ImageNet [61], and the VGG16 model has shown good performance, achieving 82% accuracy in the DSSC system [45]. Several studies have applied VGG architectures for dermatological disease diagnosis. For example, a fine-tuned VGG16 model trained on a Kaggle dataset achieved 90.1% accuracy, successfully distinguishing between visually similar skin disorders [62]. Comparative analyses have shown that while VGG19 tends to outperform other CNNs architectures such as YOLOv3 and MobileNet in terms of classification accuracy [63]. Overall, these results confirm that VGG architectures, especially VGG16 and VGG19, remain robust benchmarks for feature extraction in dermatological

image analysis.

ResNet architectures have shown high performance in dermatological image classification because of their deep residual connections, which reduce vanishing gradient issues and enable efficient feature extraction. In study [64], a ResNet50 model was trained on the HAM10000 dataset to classify seven common skin disease types, and the model achieved 91.71% accuracy, indicating an excellent balance between sensitivity and specificity in skin disease detection. Another study applies several pertained CNNs model, Xception, InceptionResNetV2, ResNet50, and VGG16, and uses two public datasets (DermNet and a skin disease classification dataset). Results indicated that ResNet50 consistently provided reliable classification performance, that mean its strong model ability and suitability for automated diagnostic systems [65]. These findings confirm that ResNet50 is a robust backbone for multi-class skin disease classification.

MobileNet architecture (MobileNetV2, MobileNetV3-Small, NAS-NetMobile) has recently gained high popularity for dermatological image classification since they are light-weight and can easily be

deployed on resource-constrained devices such as smartphones. Early work involving the usage of MobileNetV2 showed very promising results, with an accuracy of 83% on custom datasets, hence proving its potential for automated and efficient skin disease detection while minimizing computational demands [61]. Reddy, Roy, and colleagues (2023) proposed combining VGG16 and MobileNet through a transfer learning ensemble approach to increase classification accuracy while reducing computational cost. The proposed model was trained on the HAM10000 and MNIST datasets, and the results demonstrated that it outperformed conventional architectures in terms of both performance and efficiency. The accuracy has 93.75% while maintaining a lightweight and fast design.

Further research has then outlined the optimized practical usability of MobileNet in real medical applications, using datasets that contain various conditions like melanoma, seborrheic keratosis, and benign nevi, reaching accuracies of more than 95% [66]. Newer versions like MobileNetV3-Small have also been optimized for mobile skin lesion analysis. In this study on diagnosing Herpes Zoster, the model reported an accuracy of 93.5% with very good robustness to 75 types of image corruption, based on a knowledge distillation and curriculum training strategy [17]. On the other hand, researchers fine-tuned MobileNet using different learning rates and optimizers: SGD and Adam. They thus reached the maximum possible accuracy of 97.78% with the Adam optimizer at the 33rd epoch, confirming its efficiency in balancing precision and computational cost [67]. These studies position MobileNet and its variants as one of the most promising candidates for lightweight, real-time, and mobile-integrated dermatological diagnostic systems that can achieve high accuracy with their computation efficiency, thus marking a further step toward accessible and portable healthcare. Table 8 summarizes selected studies that applied Transfer Learning classifiers for skin disease classification. It highlights the dataset sources, number of images, disease classes, applied algorithms, and the reported performance results.

The TL model has proven to be an effective approach in dermatological disease classification, particularly when working with limited datasets. Models such as VGG16 and VGG19 consistently achieved high performance, above 90% in several studies [17,63], although their computational requirements remain relatively high. ResNet50 shows reliable performance across different datasets, with results above 91% on HAM10000 [64], highlighting the robustness of residual learning for medical image analysis. MobileNet with different versions (V2 and V3 Small) achieved competitive results between 83% and 97% [17,61,66,67], confirming their value as lightweight and efficient models suitable for mobile applications.

Several studies have combined TL and hybrid approaches, for example, MobileNet and VGG16, further improved accuracy to 93.75%, suggesting that integrating multiple pre-trained models enhances generalization [81]. Overall, transfer learning not only reduces the training cost and time but also enables models to leverage rich feature representations learned from large-scale datasets, making it one of the most impactful approaches in dermatology AI research.

4.3.4. Hybrid classifiers

Hybrid classifiers have gained increasing attention for improving diagnostic accuracy and enhancing generalisation by combining complementary models. For example, a hybrid DL model that integrates VGG16 and MobileNet has achieved an impressive accuracy of 94.5% across eight disease classes from the Kaggle dataset, illustrating the benefits of transfer learning ensembles in balancing performance and efficiency [66]. Another hybrid CNN model combining VGG16, MobileNet, and InceptionV3 was explored for nine-class classification, attaining 77.77% accuracy and revealing the trade-off between model depth and computational cost [82]. In more advanced architectures, a hybrid model merging DenseNet121 and EfficientNetB0 achieved 98.18% accuracy on a large 19-class custom dataset, demonstrating strong generalisation across different dermatological diseases [73].

Table 8
Summary of selected studies using transfer learning classifiers for skin disease classification.

Ref	Year	Dataset	Dataset Size	Classes	Classifiers	Model Efficiency	Result	External validation	Leakage safeguards
[68]	2022	HAM10000	1400 images	7 diseases	Custom CNN, InceptionV3, SVM, RF, KNN, DT, LR, NB, LDA	N/A	Custom CNN 95.18%	N	Not Reported
[53]	2022	DermNet	849 images	4 diseases	CNN + FedAvg	N/A	94.15%	N	Not Reported
[69]	2023	Kaggle	2400 images	8 diseases	VGG16 + MobileNet	N/A	94.5%	N	Not Reported
[70]	2023	multi-dataset	9665 images	3 diseases	CNN, LSTM, Bi-LSTM, Inception V3, VGG-16, and Xception	N/A	Xception: 99.33%	N	Not Reported
[71]	2024	Kaggle	4500 images	9 diseases	CNN VGG16, MobileNet, and InceptionV3.	N/A	proposed CNN: 77.77%	N	Not Reported
[72]	2025	Custom dataset	19,171 images	19 classes	ML models: RF, DT, SVM, KNN, LR DenseNet121 + EfficientNetB0	N/A	98.18%	N	Not Reported
[73]	2024	SCIN	10,000 images	+4 classes	FCRN + CNN	N/A	99.5%	N	Not Reported
[74]	2023	ISIC 2020	N/A	2 classes	Mask R-CNN + ResNet50	N/A	96.75%	N	Not Reported
[75]	2022	ISIC Archive	3297 images	2 classes	Hybrid Model: Deep feature extraction using ResNet50, Xception, VGG16, combined with SVM, KNN, NN, RF, LR	N/A	Xception features: 90.9%	N	Not Reported
[76]	2024	ISIC Archive + ISIC 2019-20	9200 images	2 classes	ResNet50 + VGG16 + Custom CNN + Grad-CAM	N/A	CNN + ResNet: 95%	N	Not Reported
[77]	2025	DermNet	19,500 images	23 classes	CHASHNIt (EfficientNetB7, DenseNet201, and InceptionResNetV2)	N/A	97.8%	N	Not Reported
[78]	2023	Private Acne dataset + ISIC 2019	25,331 images	8 classes	ConvNeXt ST-AFF	N/A	92.16%	N	Not Reported
[79]	2021	ISIC 2019	25,331 images	9 classes	Federated Learning Model + CNN	N/A	95.6%	N	Not Reported
[80]	2024	Kaggle + ISIC	25,010 images	5 broad categories/40 skin conditions	Xception + transfer learning models	N/A	95%	N	Not Reported

Additionally, integrating ResNet50, Xception, and VGG16 with traditional ML classifiers such as SVM, KNN, and RF improved diagnostic accuracy, with Xception-based feature extraction achieving 90.9% accuracy [75].

In a study by Muntasir et al. (2024), ResNet50 and VGG16 were utilised within custom hybrid CNN models combined with explainable AI techniques like Grad-CAM, achieving 95% accuracy and improving interpretability in skin lesion diagnostics. [73]. Combining models, such as Mask R-CNN (segmentation) and ResNet50 (classification), are hybrid models and have demonstrated greater robustness. These hybrids are reported to be approximately 96.75% classification accurate, but regarding segmentation, results are typically presented in terms of Dice/IoU and cannot be directly equated with classification accuracy [74].

Recent studies have further expanded hybridisation strategies by integrating architectures and attention mechanisms. For example, S. Anand et al. (2025) proposed CHASHNIt (composite EfficientNetB7, DenseNet201, and InceptionResNetV2) trained on 19,500 images across 23 skin disease classes, achieving 97.8% accuracy. Hao et al. (2023) developed ConvNeXt-ST-AFF, a hybrid model combining ConvNeXt and Swin Transformer with Attentional Feature Fusion (AFF) and Efficient Channel Attention (ECA) modules, using a private acne dataset and ISIC 2019 (25,331 images, eight classes), resulting in 92.16% accuracy and strong generalization across data sources. Furthermore, Hashmani et al. (2021) employed an Adaptive Federated Learning Model combined with CNN on the ISIC 2019 dataset and achieved 95.6% accuracy.

Overall, these studies confirm the growing role of hybrid models in dermatology, showing that the combination of multiple architectures can enhance robustness, improve feature extraction, and transfer more reliable skin disease classification results across different datasets and imaging conditions. The approaches, however, reflect the potential of hybrid approaches in ensuring robustness and versatility in the classification of skin disease. Table 9 summarizes selected studies that applied

Hybrid classifiers for skin disease classification.

The hybrid models showed complementary improvements over individual architectures. For example, combining VGG16 with MobileNet achieved high accuracy and lower computational costs, while the Xception model attained a high accuracy of 99.33% on 9665 images [81]. This exceptional result highlights the potential of lightweight hybrid CNN architectures for accurately classifying skin lesions. Although hybrid systems have demonstrated competitive performance in these studies, their use in dermatology remains relatively limited compared with conventional DL and transfer learning approaches, indicating a promising area for future research.

Fig. 9 illustrates a generic classification pipeline commonly adopted across the reviewed dermatological image analysis studies, highlighting how different classifier families are integrated to support automated disease diagnosis.

5. Discussion

The findings of this systematic review align closely with the broader body of literature on the application of AI in medical image analysis, while also highlighting some distinctive advances in dermatology. Earlier surveys, such as those by Litjens et al., established the potential of DL techniques for a wide range of medical imaging tasks, including classification, segmentation, and anomaly detection. The results of the current review confirm that these general trends have been translated into dermatology, where CNNs have become the foundation for automated diagnostic systems. Similar to previous research, CNNs consistently outperformed traditional ML approaches by reducing their dependency on handcrafted features and demonstrating higher accuracy across diverse datasets.

Overall, when compared with previous literature, this review highlights both continuity and innovation. The continuity is evident in the reliance on CNN-based architectures and the use of large-scale datasets

Table 9
Summary of selected studies using hybrid classifiers for skin disease classification.

Ref	Year	Dataset	Dataset Size	Classes	Classifiers	Model Efficiency	Result	External validation	Leakage safeguards
[68]	2022	HAM10000	1400 images	7 diseases	Custom CNN, InceptionV3, SVM, RF, KNN, DT, LR, NB, LDA	N/A	Custom CNN 95.18%	N	Not Reported
[53]	2022	DermNet	849 images	4 diseases	CNN + FedAvg	N/A	94.15%	N	Not Reported
[69]	2023	Kaggle	2400 images	8 diseases	VGG16 + MobileNet	N/A	94.5%	N	Not Reported
[70]	2023	N/A	9665 images	3 diseases	CNN, LSTM, Bi-LSTM, Inception V3, VGG-16, and Xception	N/A	Xception: 99.33%	N	Not Reported
[71]	2024	Kaggle	4500 images	9 diseases	CNN VGG16, MobileNet, and InceptionV3. ML models: RF, DT, SVM, KNN, LR	N/A	proposed CNN: 77.77%	N	Not Reported
[72]	2025	Custom dataset	19,171 images	19 classes	DenseNet121 + EfficientNetB0	N/A	98.18%	N	Not Reported
[73]	2024	SCIN	10,000 images	+4 classes	FCRN + CNN	N/A	99.5%	N	Not Reported
[74]	2023	ISIC 2020	N/A	2 classes	Mask R-CNN + ResNet50	N/A	96.75%	N	Not Reported
[75]	2022	ISIC Archive	3297 images	2 classes	Hybrid Model: Deep feature extraction using ResNet50, Xception, VGG16, combined with SVM, KNN, NN, RF, LR	N/A	Xception features: 90.9%	N	Not Reported
[76]	2024	ISIC Archive + ISIC 2019-20	9200 images	2 classes	ResNet50 + VGG16 + Custom CNN + Grad-CAM	N/A	CNN + ResNet: 95%	N	Not Reported
[77]	2025	DermNet	19,500 images	23 classes	CHASHNIt (EfficientNetB7, DenseNet201, and InceptionResNetV2)	N/A	97.8%	N	Not Reported
[78]	2023	Private Acne dataset + ISIC 2019	25,331 images	8 classes	ConvNeXt ST-AFF	N/A	92.16%	N	Not Reported
[79]	2021	ISIC 2019	25,331 images	9 classes	Federated Learning Model + CNN	N/A	95.6%	N	Not Reported
[80]	2024	Kaggle + ISIC	25,010 images	5 broad categories/40 skin conditions	Xception + transfer learning models	N/A	95%	N	Not Reported

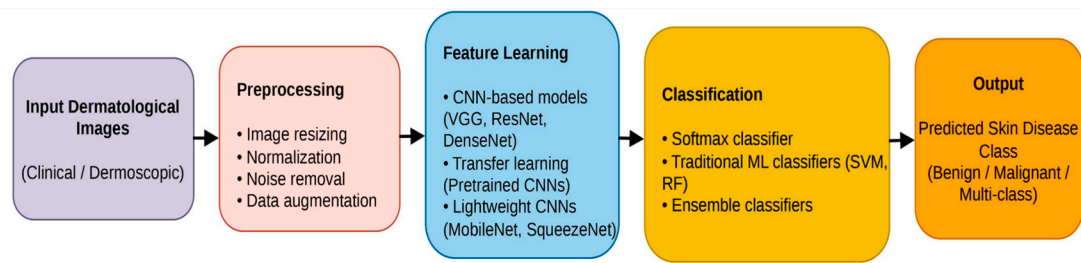


Fig. 9. Generic pipeline for image-based dermatological disease classification.

to advance classification accuracy. The innovation lies in the adoption of lightweight and hybrid models, the incorporation of transfer learning, and the increased attention to real-world deployment issues such as efficiency, accessibility, and fairness. Together, these findings suggest that dermatological AI research is moving beyond proof-of-concept studies and toward scalable, clinically relevant solutions that build upon, but also expand, the foundations laid by earlier medical imaging research.

This section displays the analysis and discussion of the findings obtained from this SR. The results provide answers to the research questions that were formulated, which are as follows.

RQ1: What methodological trends (ML, DL, transfer learning, and hybrid models) characterize dermatological skin disease classification studies published between 2018 and 2025?

The shift from traditional ML techniques and handcrafted features towards DL with CNNs, transfer learning, and hybrid models has been well documented. Earlier classical ML research, which used colour and texture descriptors with SVMs, KNNs, and RFs, produced decent results with accuracy rates that varied, e.g., LDA/SVM on clinician-labelled images achieved 82.58% accuracy, KNN/RF with a 21k image benchmark reached around 95%, and a machine learning ensemble on five classes achieved 97.33% [29]. However, these studies relied on feature engineering, which led to models that did not generalize well. Most recent work has been dominated by DL, with clinical settings using EfficientNet frameworks achieving 94.8% accuracy [43], modified ResNet50 performing classification tasks with 98.93% accuracy [37], and lightweight CNNs for dermatology delivering remarkable performance [42,44]. Various techniques to improve training and generalisation of CNNs, such as data augmentation and generative adversarial networks, are well documented [41,83]. Hybrid models, incorporating layers for segmentation and classification, as well as stacked detection-classification architectures, such as a combination of YOLOv8 and CNN, offer improvements in model robustness and end-to-end usability [50,56,81]. The prevalence of DL using transfer learning can be attributed to the automatic learning of features, as suggested by multiple sources [37,38,40,43,50,84,85].

RQ2: What datasets have been employed in dermatology classification studies?

Most studies utilise publicly accessible dermoscopy and clinical repositories as benchmarks, such as HAM (10,015 images, 7 classes) [23], ISIC challenges/archives (2016-2020) [4], and DermNet collections [16], as well as the XiangyaDerm dataset, which contains over 100,000 images and 541 conditions [24]. Additionally, recent research has introduced newer large-scale resources like the DERM12345 dataset, offering a comprehensive and diverse collection of dermatological images across multiple disease categories, supporting multi-class skin disease classification studies [7]. Although these datasets facilitate some comparability, studies have reported fairness concerns regarding performance gaps with visually distinct, diverse sets [25]. Despite

documentation of class imbalance, underrepresentation of darker skin tones, and the inclusion of more recent multi-tone datasets (e.g., Fitzpatrick-style subsets, where available), various studies employ these datasets alongside traditional benchmarks [20,23,25]. However, most of these benchmark datasets remain predominantly image-centric, with limited structured clinical metadata such as patient history, lesion evolution, anatomical site, or demographic attributes. This reliance on visual data alone may restrict clinical interpretability and real-world diagnostic robustness, underscoring the potential advantage of integrating multimodal data sources into future dermatological AI systems.

RQ3: What validation techniques are used to evaluate the performance of the proposed models?

Most publications assess accuracy as the main metric within train/validation/test splits or k-fold cross-validation (usually 5 to 10 folds), along with increasing usage of the F1 score, sensitivity/specificity, and AUC to tackle imbalance and minority-class detection. For instance, MobileNet pipelines demonstrate accuracy varying from 85% to 97% depending on the dataset and setup [5,66,67] with EfficientNet frameworks presenting accuracy and F1 score (94.8%) in clinical settings with multiple classes [43], attention-augmented CNNs reported accuracy and performed confusion-level assessments [50], and studies on ResNet50 presented multi-class accuracy on standard splits of around 92% and above [64]. In detection-classification systems such as YOLOv8 + CNN, the detection performance is evaluated based on precision, recall and mAP, and the classification step is measured based on accuracy, F1 score and AUC. These measures must not be combined in tasks [5,43,50,56,64].

RQ4: How do lightweight architectures, particularly MobileNet, perform compared with other DL models in terms of diagnostic accuracy and computational efficiency?

Light CNNs maintain impressive accuracy while utilizing considerably less computational power, thus proving their suitability for deployment in mobile devices and point-of-care situations. MobileNet and its variants achieve roughly 85% accuracy on 9-class Kaggle data [5,86], nearly 95% on tri-class HAM10000 data [66], 97.78% on 6-class tasks [67,87], and after fine-tuning, notable fairness-oriented performance on tone-stratified cyanosis detection [48,88]. Other compact designs, such as SqueezeNet, also achieve stunning results on cancer/lesion classification while maintaining parameter minimalism [44]. MobileNet-RseSK advances feature calibration through residual squeeze and excitation [39], which further enhances performance. While heavier backbones such as ResNet, VGG, and EfficientNet reach higher performance on some benchmarks (e.g., 94.8% EfficientNet clinic, 98.93 on modified ResNet50), it does come with higher computational costs [37,43]. To further illustrate the relationship between computational complexity and diagnostic performance, Fig. 9 presents a comparative visualization of model size (in million parameters) versus reported accuracy on the HAM10000 benchmark dataset. This representation provides an intuitive overview of the efficiency-accuracy trade-off across

lightweight and larger architectures.

As shown in Fig. 10, lightweight architectures such as MobileNet-based models and LWCNN achieve competitive accuracy while maintaining substantially lower parameter counts compared to heavier backbones such as ResNet50. This observation reinforces the suitability of compact models for real-world and mobile dermatology applications, where computational resources may be limited. In particular, MobileNet's depthwise separable convolution design enables strong accuracy-per-FLOP performance, and its native support within mobile deployment frameworks such as TensorFlow Lite (TF-Lite) and CoreML further strengthens its practical suitability for resource-constrained and point-of-care applications. Hybrid stacks, which combine lightweight backbones with segmentation or attention modules, considerably reduce and maintain any inefficiency in accuracy [50,56,81]. While MobileNet is highlighted due to its favorable deployment characteristics and widespread adoption in dermatological AI studies, other lightweight architectures such as EfficientNet variants, SqueezeNet, and NASNetMobile also demonstrate competitive efficiency-performance trade-offs across multiple datasets [89]. Therefore, the choice of lightweight backbone remains context-dependent and influenced by dataset characteristics, class imbalance, and deployment constraints. Overall, these findings collectively address RQ4 and situate lightweight architectures within the broader evolution of dermatological AI research.

For all four research questions, the field has made a significant shift towards CNN-based DL techniques that incorporate transfer learning and selective hybrids. Progress has been made with public datasets (HAM10000/ISIC/XiangyaDerm), but issues with dataset imbalance and diversity in skin tone, limited use of external validation, and lack of explicitly reported data leakage safeguards remain [20,25]. Evaluation methodologies extending beyond accuracy to incorporate F1 or AUC are more common. For practical deployment that remains competitively performant, lightweight architectures such as MobileNets, SqueezeNets, and efficient variants are emerging as frontrunners [5,39,44,66,67] while advanced backbones and attention modules are useful for resource-intensive tasks to maximise performance [37,43,50,83–85,90]. The era of AI-powered dermatological diagnosis was marked by the pioneering study of Esteva et al. [91], who showed that the dermatologist's level of skin cancer classification could be performed with deep convolutional neural networks. This was also supported by Haenssle et al. [92], who demonstrated that a CNN was able to outperform most dermatologists in melanoma recognition, which made AI a clinically viable decision-support technology. Artificial intelligence methods of analysing dermatological images have been reviewed several times, but

mainly with a particular emphasis on deep learning networks, widely used datasets and diagnostic performance testing [93,94]. The works offer a general overview of the current trends in methodology in the area of dermatology AI, and create the background of research in terms of image-based skin diseases classification.

5.1. Future directions: toward multi-modal dermatology AI

While this review primarily focuses on image-based (single-modal) dermatological AI systems, the analysis presented in this study highlights that most benchmark datasets remain predominantly image-centric, with limited structured clinical metadata. In real-world dermatological practice, diagnosis often relies not only on visual inspection but also on contextual clinical details such as patient history, lesion duration, anatomical site, and demographic characteristics. This review is written with a focus on image-based (single-modal) dermatological AI systems, new multimodal techniques that combine clinical metadata, patient history, or text information could potentially add to the strength of diagnosis. Future research should systematically evaluate whether multimodal fusion architectures provide clinically meaningful improvements over high-performing single-modal convolutional neural networks. Such approaches may also contribute to improved fairness and generalizability across diverse patient populations, particularly in addressing challenges related to skin tone variation and demographic imbalance.

As summarised in Table 10, each of the reviewed techniques has obvious trade-offs between classification effectiveness, computational complexity, and deployment viability. Deep learning and transfer learning methods have better accuracy, but they are in many cases more resource intensive in annotating large datasets and computational resources. Conversely, traditional machine learning and lightweight architectures are more efficient, interpretable and favorable to resource-constrained environments at the expense of a minor loss in peak performance. These trade-offs emphasise the need to choose AI techniques depending on clinical and deployment limitations.

Other potential directions in dermatological image analysis are fairness-aware modeling with varied skin-tone modeling, external validation using a heterogeneous dataset, and deployable lightweight architectures. The model interpretability and uncertainty estimation is also to be studied further and improve clinical reliability and adoption.

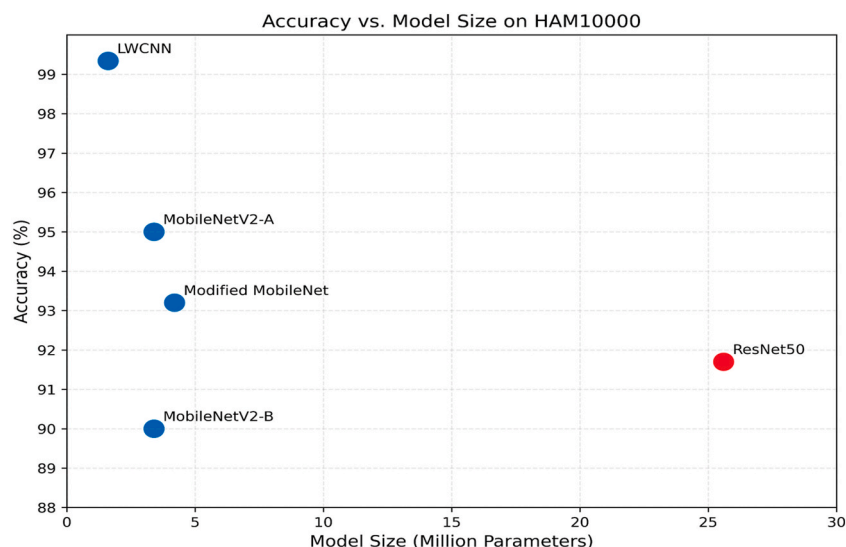


Fig. 10. Comparison of model accuracy versus computational complexity for selected architectures on the HAM10000 dataset.

Table 10
Comparative analysis of techniques used in dermatological disease classification.

Approach	Merits	Demerits
Traditional ML (SVM, KNN, RF, NB)	Low computational cost; interpretable; effective on small datasets	Requires handcrafted features; limited scalability; poor generalization
Deep Learning (CNNs)	Automatic feature learning; high accuracy; robust to visual variability	Data-hungry; high computational and memory cost
Transfer Learning	Faster convergence; improved performance on limited data; reuse of pretrained knowledge	Domain mismatch risk; reduced interpretability
Lightweight CNNs (MobileNet, SqueezeNet)	High accuracy-per-FLOP; low latency; suitable for mobile/edge deployment	Slightly lower peak accuracy than heavy models
Hybrid Models (CNN + ML/multi-CNN)	Improved robustness; complementary feature utilization	Increased complexity; harder to deploy and tune
Detection/Segmentation-based Models	Lesion localization improves interpretability and diagnosis	Additional annotation cost; higher training complexity

6. Conclusions

The evidence collected in this review demonstrates the considerable advancements employed in DL and lightweight architectures concerning the classification of dermatological diseases. Still, future studies need to prioritize the gaps worth addressing, particularly diversity within the datasets, clinical validation, and real-world application. The fairness and algorithmic bias of these systems hinge on the expansion of datasets to include more class-balanced and representative varying skin tones. The development of international collaborative databases could increase the likelihood of these datasets containing larger and more representative collections of the necessary images. Though the existing models have shown strong performance in controlled research settings, most models have seen very limited deployment in clinical practice. Future work should include prospective studies as well as testing models in varied clinical workflows to assess the reliability, usability, and ethical implications. This is necessary to develop trust within the clinician community and ultimately, the integration of AI-derived tools in routine medical practice.

Future research should also focus on achieving the optimal balance between accuracy and efficiency. Lightweight models like MobileNet, EfficientNet, and SqueezeNet have demonstrated that high accuracy can be achieved with lower computational requirements, which is ideal for mobile and point-of-care situations. These innovations can lower the costs associated with diagnoses, allow specialised dermatological services to be more widely available, and facilitate the early detection of dangerous conditions like melanoma. Results show DL, and the CNN specifically, as the most powerful, as well as outperforming classical machine learning models, which rely heavily on handcrafted features. Lightweight architectures, in particular, demonstrate great potential in achieving diagnostic accuracy and efficiency in computation.

The review also highlights the crucial role of publicly available datasets such as ISIC, HAM10000, DermNet, and Xiangya Derm, which have enabled significant advancements in model development. At the same time, persistent challenges remain, particularly issues of dataset imbalance and underrepresentation of darker skin tones, which limit the generalizability and fairness of existing systems. While most studies reported high levels of accuracy, the overreliance on retrospective datasets and limited clinical validation raises concerns regarding the real-world applicability of these models.

Despite these limitations, the overall trajectory of research indicates a strong move toward efficiency-driven designs, hybrid approaches, and transfer learning techniques that enhance robustness and adaptability. Future research should continue to address the identified gaps, with

particular emphasis on improving dataset diversity, strengthening clinical validation, and aligning with ethical and regulatory standards.

In conclusion, AI-driven models, especially lightweight CNN architectures, represent a transformative opportunity in dermatological diagnostics. By enhancing diagnostic accuracy, reducing costs, and expanding accessibility, these technologies can complement clinical expertise and support timely, reliable, and scalable healthcare delivery.

CRediT authorship contribution statement

Bodoor H.S. Aljohani: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Wael M.S. Yafooz:** Writing – review & editing, Validation, Supervision, Project administration, Methodology. **Abdullah Alsaeedi:** Writing – review & editing, Validation, Supervision.

Ethical approval

This study is a systematic review of previously published research and publicly available datasets. No new human subjects were recruited, and no identifiable personal data were collected or analyzed. Therefore, formal ethical approval was not required.

Declaration of generative AI and AI-assisted technologies

No generative AI tools were used in the preparation of this manuscript. All sections were written, analyzed, and verified completely by the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.array.2026.100797>.

Appendix A. Detailed Search Strategy:

The following verbatim search strings were used during database querying. All searches were conducted between 14 April 2025 and 25 October 2025. The search process began with exploratory keyword-based queries in Google Scholar to identify commonly used terminology in dermatological disease classification research. Initial search phrases included: “skin disease classification”, “dermatology classification”, “deep learning in skin disease classification”, “computer vision in skin disease classification”, “machine learning in skin disease classification”, “hybrid model for skin disease detection”. As the review progressed, structured Boolean combinations (using AND/OR operators) were applied across databases to refine and narrow the results. Search syntax was adapted to each database's indexing rules (IEEE Xplore, PubMed, Web of Science, ACM Digital Library, SpringerLink, ScienceDirect, and Google Scholar). Database-specific adjustments (e.g., field tags and indexing syntax) were implemented where required to align with each platform's search interface. Filters were applied to limit results to English-language peer-reviewed publications published between 2018 and 2025. All potentially relevant studies were screened by

reviewing titles and abstracts prior to full-text assessment.

Data availability

All data used in this systematic review were obtained from publicly available dermatology datasets and published research articles. The supporting data underlying this study have been made publicly available in the Mendeley Data repository (DOI: <https://doi.org/10.17632/b8h9jbc6mn.1>).

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